

Volatility Spread Exercise

1. In option trading a trader is said to have sold, or is short, a position if the entire trade results in a credit. A trader is said to have bought, or is long, a position if the entire trade results in a debit.

For each spread position below, identify its name or type, and, where applicable, whether the spread is “long” or “short”. Then, assuming that each position is approximately delta neutral, what is the initial sign (+ or -) of the gamma, theta, and vega?

	<u>name</u> <u>or type</u>	<u>initial</u> <u>gamma</u>	<u>initial</u> <u>theta</u>	<u>initial</u> <u>vega</u>
a.)	+1 January 70 call +1 January 70 put			
b.)	+1 February 65 call -1 March 65 call			
c.)	+1 February 70 put -3 February 60 puts			
d.)	+1 March 50 call -2 March 55 calls +1 March 60 call			

Volatility Spread Exercise

1. continued	<u>name</u> <u>or type</u>	<u>initial</u> <u>gamma</u>	<u>initial</u> <u>theta</u>	<u>initial</u> <u>vega</u>
e.)	+1 March 65 call +1 March 70 put			
f.)	+1 June 100 put -1 April 100 put			
g.)	+2 April 120 calls -1 April 110 call			
h.)	-1 June 105 put -1 June 120 call			
i.)	-1 July 25 put +1 July 30 put +1 July 35 put -1 July 40 put			
j.)	+1 July 35 put -1 August 35 put			

Volatility Spread Exercise

1. continued	<u>name</u> <u>or type</u>	<u>initial</u> <u>gamma</u>	<u>initial</u> <u>theta</u>	<u>initial</u> <u>vega</u>
k.)	+1 July 40 put +1 July 50 call			
l.)	+1 August 40 call -2 August 45 calls			
m.)	+3 November 1850 puts -2 November 1900 puts			
n.)	+1 December 2000 call -1 October 2000 call			
o.)	-1 November 1900 call +1 November 1950 call +1 November 2000 call			
p.)	-1 December 2000 put +1 December 2025 put +1 December 2050 call -1 December 2075 call			

Volatility Spread Exercise

2. Option contracts are trading at the following prices with the given delta values:

	<u>call</u> <u>price</u>	<u>call</u> <u>delta</u>	<u>put</u> <u>price</u>	<u>put</u> <u>delta</u>
May 75	7.94	88	.43	-12
May 80	4.19	66	1.64	-34
May 85	1.77	38	4.18	-62
May 90	.59	17	7.97	-83
July 75	9.16	83	.96	-17
July 80	5.68	66	2.40	-34
July 85	3.17	46	4.81	-54
July 90	1.59	28	8.15	-72

Using the above table, what are the prices and delta values if you “buy” each of the following spreads?

Volatility Spread Exercise

2. continued

- a.) July 85 straddle
- b.) July / May 80 call calendar spread
- c.) May 75 / 85 1x2 call spread
- d.) July 80 / 85 / 90 put butterfly
- e.) July 75 / 90 strangle
- f.) July 75 / 85 1x3 put spread
- g.) May 80 / 85 “guts” strangle

Volatility Spread Exercise

2. continued

- h.) May 75 / 80 / 85 / 90 put condor
- i.) May 75 / 80 / 85 / 90 iron condor
- j.) July / May 85 put calendar spread
- k.) July 75 / 85 / 90 call ladder
- l.) July 80 / 85 / 90 call butterfly
- m.) July 80 / 85 / 90 iron butterfly
- n.) May 80 / 85 / 90 put ladder

Volatility Spread Exercise

3. Below are several different stock option positions with some suggested changes in market conditions. If the underlying stock is currently trading at 80, for each change in market conditions, is the position making money (+), losing money (-), or staying about the same (0)? For ratio spreads (the number of long and short market contracts are unequal), assume that the spread is approximately delta neutral. Assume that dividends are paid quarterly.

a.)	+1 March 90 call +1 March 80 put	the stock price <u>rises sharply</u>	time passes with <u>no movement</u>	implied volatility <u>rises</u>
b.)	+1 March 85 put -3 March 75 puts	the stock price <u>rises sharply</u>	time passes with <u>no movement</u>	implied volatility <u>rises</u>
c.)	+1 March 80 put -1 June 80 put	interest rates <u>rises sharply</u>	time passes with <u>no movement</u>	implied volatility <u>falls</u>

Volatility Spread Exercise

3. continued

d.)	+1 June 75 put -2 June 80 puts +1 June 85 put	the stock price <u>falls sharply</u>	time passes with <u>no movement</u>	implied volatility <u>falls</u>
e.)	+1 March 80 call +1 March 80 put	the stock price <u>rises sharply</u>	time passes with <u>no movement</u>	implied volatility <u>rises</u>
f.)	-1 March 80 call +1 June 80 call	the stock price <u>rises sharply</u>	the dividend <u>is reduced</u>	implied volatility <u>falls</u>

Volatility Spread Exercise

3. continued

	the stock price <u>falls sharply</u>	time passes with <u>no movement</u>	implied volatility <u>rises</u>
g.) +1 June 75 call -3 June 85 calls			
	interest rates <u>fall sharply</u>	the dividend <u>is increased</u>	implied volatility <u>falls</u>
h.) -1 March 80 put +1 June 80 put			
	the stock price <u>rises sharply</u>	time passes with <u>no movement</u>	implied volatility <u>falls</u>
i.) -1 March 70 call +1 March 75 call +1 March 85 call -1 March 90 call			

Volatility Spread Exercise

4. With the underlying contract trading close to 50, in each group of spreading strategies choose both the strategy which will have the highest price and the strategy which will have the lowest price in the marketplace. Assume that there are no interest or dividend considerations, and that April options expire before June options.

a.) +1 April 40 call / +1 April 40 put
 +1 April 45 call / +1 April 45 put
 +1 April 50 call / +1 April 50 put
 +1 April 55 call / +1 April 55 put

b.) +1 June 45 put / -1 April 45 put
 +1 June 50 put / -1 April 50 put
 +1 June 55 put / -1 April 55 put
 +1 June 60 put / -1 April 60 put

c.) +1 May 40 put / +1 May 60 call
 +1 May 45 put / +1 May 55 call
 +1 July 40 put / +1 July 60 call
 +1 July 45 put / +1 July 55 call

Volatility Spread Exercise

5. continued

d.) +1 April 55 call / +1 April 55 put
+1 April 60 call / +1 April 60 put
+1 April 50 call / +1 April 50 put
+1 April 45 call / +1 April 45 put

e.) +1 April 50 call / +1 April 50 put
+1 June 50 call / +1 June 50 put
+1 April 55 call / +1 April 55 put
+1 June 55 call / +1 June 55 put

f.) +1 April 45 call / -2 April 50 calls / +1 April 55 call
+1 April 40 put / -2 April 50 puts / +1 April 60 put
+1 April 40 call / -2 April 45 calls / +1 April 50 call
+1 April 45 put / -2 April 50 puts / +1 April 55 put

Volatility Spread Exercise

5. continued

g.) +1 April 40 call / -2 April 45 calls / +1 April 50 call
+1 June 50 put / -2 June 55 puts / +1 June 60 put
+1 April 40 put / -2 April 50 puts / +1 April 60 put
+1 June 40 call / -2 June 50 calls / +1 June 60 call

e.) +1 June 50 call / -1 April 50 call
+1 June 55 call / -1 April 55 call
+1 June 45 put / -1 April 45 put
+1 June 40 put / -1 April 40 put

Bull and Bear Exercise

For each of the following groups of spreads you are given an underlying price, a directional outlook (bullish or bearish), and an opinion about implied volatility (unusually high or unusually low). Given this information, choose the best spread from among the four possible choices. Assume that all options expire at the same time.

	<u>underlying price</u>	<u>directional outlook</u>	<u>implied volatility</u>	<u>spread choices</u>
a.)	90	bullish	unusually high	long a 95 put / short a 90 put long an 85 call / short a 90 call long a 90 put / short an 85 put long a 90 call / short an 85 call
b.)	55	bearish	unusually low	long a 55 put / short a 50 put long a 55 call / short a 60 put long a 50 call / short a 55 put long a 60 call / short a 55 call
c.)	127	bearish	unusually high	long a 135 put / short a 140 put long a 120 call / short a 130 call long a 115 put / short a 125 put long a 135 call / short a 125 call

Bull and Bear Exercise

	<u>underlying price</u>	<u>directional outlook</u>	<u>implied volatility</u>	<u>spread choices</u>
d.)	248	bullish	unusually low	long a 250 call / short a 230 call <i>long a 250 call / short a 270 call</i> long a 250 put / short a 230 put long a 230 put / short a 250 put

Volatility Spread Exercise (answers)

1. In option trading a trader is said to have sold, or is short, a position if the entire trade results in a credit. A trader is said to have bought, or is long, a position if the entire trade results in a debit.

For each spread position below, identify its name or type, and, where applicable, whether the spread is “long” or “short”. Then, assuming that each position is approximately delta neutral, what is the initial sign (+ or -) of the gamma, theta, and vega?

	<u>name or type</u>	<u>initial gamma</u>	<u>initial theta</u>	<u>initial vega</u>
a.) +1 January 70 call +1 January 70 put	<i>long straddle</i>	+	-	+
b.) +1 February 65 call -1 March 65 call	<i>short call calendar spread</i>	+	-	-
c.) +1 February 70 put -3 February 60 puts	<i>put ratio spread</i>	-	+	-
d.) +1 March 50 call -2 March 55 calls +1 March 60 call	<i>long call butterfly</i>	-	+	-

Volatility Spread Exercise (answers)

1. continued	name or type	initial gamma	initial theta	initial vega
e.) +1 March 65 call +1 March 70 put	<i>long strangle (guts)</i>	+	-	+
f.) +1 June 100 put -1 April 100 put	<i>long put calendar spread</i>	-	+	+
g.) +2 April 120 calls -1 April 110 call	<i>call ratio spread</i>	+	-	+
h.) -1 June 105 put -1 June 120 call	<i>short strangle</i>	-	+	-
i.) -1 July 25 put +1 July 30 put +1 July 35 put -1 July 40 put	<i>short put condor</i>	+	-	+
j.) +1 July 35 put -1 August 35 put	<i>short put calendar spread</i>	+	-	-

Volatility Spread Exercise (answers)

1. continued	name or type	initial gamma	initial theta	initial vega
k.) +1 July 40 put +1 July 50 call	<i>long strangle</i>	+	-	+
l.) +1 August 40 call -2 August 45 calls	<i>call ratio spread</i>	-	+	-
m.) +3 November 1850 puts -2 November 1900 puts	<i>put ratio spread</i>	+	-	+
n.) +1 December 2000 call -1 October 2000 call	<i>long call calendar spread</i>	-	+	+
o.) -1 November 1900 call +1 November 1950 call +1 November 2000 call	<i>short call ladder</i>	+	-	+
p.) -1 December 2000 put +1 December 2025 put +1 December 2050 call -1 December 2075 call	<i>long iron condor</i>	+	-	+

Volatility Spread Exercise (answers)

2. Option contracts are trading at the following prices with the given delta values:

	<u>call</u> <u>price</u>	<u>call</u> <u>delta</u>	<u>put</u> <u>price</u>	<u>put</u> <u>delta</u>
May 75	7.94	88	.43	-12
May 80	4.19	66	1.64	-34
May 85	1.77	38	4.18	-62
May 90	.59	17	7.97	-83
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July 80	5.68	66	2.40	-34
July 85	3.17	46	4.81	-54
July 90	1.59	28	8.15	-72

Using the above table, what are the prices and delta values if you “buy” each of the following spreads?

Volatility Spread Exercise (answers)

2. continued

- | | | |
|-----|------------------------------------|---|
| a.) | July 85 straddle | $\text{price} = 3.17 + 4.81 = \mathbf{7.98}$
$\text{delta} = 46 - 54 = \mathbf{-8}$ |
| b.) | July / May 80 call calendar spread | $\text{price} = 5.68 + 4.19 = \mathbf{1.49}$
$\text{delta} = 66 - 66 = \mathbf{0}$ |
| c.) | May 75 / 85 1x2 call spread | $\text{price} = 7.94 - (2 \times 1.77) = \mathbf{4.40}$
$\text{delta} = 88 - (2 \times 38) = \mathbf{+12}$ |
| d.) | July 80 / 85 / 90 put butterfly | $\text{price} = 2.40 + 8.15 - (2 \times 4.81) = \mathbf{.93}$
$\text{delta} = -34 - 72 - (2 \times -54) = \mathbf{+2}$ |
| e.) | July 75 / 90 strangle | $\text{price} = .96 + 1.59 = \mathbf{2.55}$
$\text{delta} = -17 + 28 = \mathbf{+11}$ |
| f.) | July 75 / 85 1x3 put spread | $\text{price} = 4.81 - (3 \times .96) = \mathbf{1.93}$
$\text{delta} = -54 - (3 \times -17) = \mathbf{-3}$ |
| g.) | May 80 / 85 “guts” strangle | $\text{price} = 4.19 + 4.18 = \mathbf{8.37}$
$\text{delta} = 66 - 62 = \mathbf{+4}$ |

Volatility Spread Exercise (answers)

2. continued

- | | | |
|-----|-----------------------------------|---|
| h.) | May 75 / 80 / 85 / 90 put condor | price = $(7.97 + .43) - (4.18 + 1.64) = \mathbf{2.58}$
delta = $(62 + 34) - (83 + 12) = \mathbf{+1}$ |
| i.) | May 75 / 80 / 85 / 90 iron condor | price = $(1.64 + 1.77) - (.43 + .59) = \mathbf{2.39}$
delta = $(38 - 34) - (17 - 12) = \mathbf{-1}$ |
| j.) | July / May 85 put calendar spread | price = $4.81 - 4.18 = \mathbf{.63}$
delta = $-54 + 62 = \mathbf{+8}$ |
| k.) | July 75 / 85 / 90 call ladder | price = $9.16 - (3.17 + 1.59) = \mathbf{4.40}$
delta = $83 - (46 + 28) = \mathbf{+9}$ |
| l.) | July 80 / 85 / 90 call butterfly | price = $(5.68 + 1.59) - 2 \times 3.17 = \mathbf{.93}$
delta = $(66 + 28) - 2 \times 46 = \mathbf{+2}$ |
| m.) | July 80 / 85 / 90 iron butterfly | price = $(3.17 + 4.81) - (1.59 + 2.40) = \mathbf{3.99}$
delta = $(-34 - 72) + (2 \times 54) = \mathbf{+2}$ |
| n.) | May 80 / 85 / 90 put ladder | price = $7.97 - (4.18 + 1.64) = \mathbf{2.15}$
delta = $-83 + (62 + 34) = \mathbf{+13}$ |

Volatility Spread Exercise (answers)

3. Below are several different stock option positions with some suggested changes in market conditions. If the underlying stock is currently trading at 80, for each change in market conditions, is the position making money (+), losing money (-), or staying about the same (0)? For ratio spreads (the number of long and short market contracts are unequal), assume that the spread is approximately delta neutral. Assume that dividends are paid quarterly.

	the stock price <u>rises sharply</u>	time passes with <u>no movement</u>	implied volatility <u>rises</u>
a.) +1 March 90 call +1 March 80 put	+	-	+
	the stock price <u>rises sharply</u>	time passes with <u>no movement</u>	implied volatility <u>rises</u>
b.) +1 March 85 put -3 March 75 puts	-	+	-
	interest rates <u>rises sharply</u>	time passes with <u>no movement</u>	implied volatility <u>falls</u>
c.) +1 March 80 put -1 June 80 put	+	-	+

Volatility Spread Exercise (answers)

3. continued

		the stock price <u>falls sharply</u>	time passes with <u>no movement</u>	implied volatility <u>falls</u>
d.)	+1 June 75 put -2 June 80 puts +1 June 85 put	-	+	+
		the stock price <u>rises sharply</u>	time passes with <u>no movement</u>	implied volatility <u>rises</u>
e.)	+1 March 80 call +1 March 80 put	+	-	+
		the stock price <u>rises sharply</u>	the dividend <u>is reduced</u>	implied volatility <u>falls</u>
f.)	-1 March 80 call +1 June 80 call	-	+	-

Volatility Spread Exercise (answers)

3. continued

		the stock price <u>falls sharply</u>	time passes with <u>no movement</u>	implied volatility <u>rises</u>
g.)	+1 June 75 call -3 June 85 calls	-	+	-
		interest rates <u>fall sharply</u>	the dividend <u>is increased</u>	implied volatility <u>falls</u>
h.)	-1 March 80 put +1 June 80 put	+	+	-
		the stock price <u>rises sharply</u>	time passes with <u>no movement</u>	implied volatility <u>falls</u>
i.)	-1 March 70 call +1 March 75 call +1 March 85 call -1 March 90 call	+	-	-

Volatility Spread Exercise (answers)

4. With the underlying contract trading close to 50, in each group of spreading strategies choose both the strategy which will have the highest price and the strategy which will have the lowest price in the marketplace. Assume that there are no interest or dividend considerations, and that April options expire before June options.

- | | | |
|-----|------------------------------------|----------------|
| a.) | +1 April 40 call / +1 April 40 put | <i>highest</i> |
| | +1 April 45 call / +1 April 45 put | |
| | +1 April 50 call / +1 April 50 put | <i>lowest</i> |
| | +1 April 55 call / +1 April 55 put | |
| b.) | +1 June 45 put / -1 April 45 put | |
| | +1 June 50 put / -1 April 50 put | <i>highest</i> |
| | +1 June 55 put / -1 April 55 put | |
| | +1 June 60 put / -1 April 60 put | <i>lowest</i> |
| c.) | +1 May 40 put / +1 May 60 call | <i>lowest</i> |
| | +1 May 45 put / +1 May 55 call | |
| | +1 July 40 put / +1 July 60 call | |
| | +1 July 45 put / +1 July 55 call | <i>highest</i> |

Volatility Spread Exercise (answers)

5. continued

- | | | |
|-----|--|---------------------------------|
| d.) | +1 April 55 call / +1 April 55 put
+1 April 60 call / +1 April 60 put
+1 April 50 call / +1 April 50 put
+1 April 45 call / +1 April 45 put | <i>highest</i>
<i>lowest</i> |
| e.) | +1 April 50 call / +1 April 50 put
+1 June 50 call / +1 June 50 put
+1 April 55 call / +1 April 55 put
+1 June 55 call / +1 June 55 put | <i>lowest</i>
<i>highest</i> |
| f.) | +1 April 45 call / -2 April 50 calls / +1 April 55 call
+1 April 40 put / -2 April 50 puts / +1 April 60 put
+1 April 40 call / -2 April 45 calls / +1 April 50 call
+1 April 45 put / -2 April 50 puts / +1 April 55 put | <i>highest</i>
<i>lowest</i> |

Volatility Spread Exercise (answers)

5. continued

- | | | |
|-----|---|----------------|
| g.) | +1 April 40 call / -2 April 45 calls / +1 April 50 call | <i>lowest</i> |
| | +1 June 50 put / -2 June 55 puts / +1 June 60 put | |
| | +1 April 40 put / -2 April 50 puts / +1 April 60 put | <i>highest</i> |
| | +1 June 40 call / -2 June 50 calls / +1 June 60 call | |
| | | |
| e.) | +1 June 50 call / -1 April 50 call | <i>highest</i> |
| | +1 June 55 call / -1 April 55 call | |
| | +1 June 45 put / -1 April 45 put | |
| | +1 June 40 put / -1 April 40 put | <i>lowest</i> |

Bull and Bear Exercise (answers)

For each of the following groups of spreads you are given an underlying price, a directional outlook (bullish or bearish), and an opinion about implied volatility (unusually high or unusually low). Given this information, choose the best spread from among the four possible choices. Assume that all options expire at the same time.

	<u>underlying price</u>	<u>directional outlook</u>	<u>implied volatility</u>	<u>spread choices</u>
a.)	90	bullish	unusually high	long a 95 put / short a 90 put long an 85 call / short a 90 call long a 90 put / short an 85 put long a 90 call / short an 85 call
b.)	55	bearish	unusually low	long a 55 put / short a 50 put long a 55 call / short a 60 put long a 50 call / short a 55 put long a 60 call / short a 55 call
c.)	127	bearish	unusually high	long a 135 put / short a 140 put long a 120 call / short a 130 call long a 115 put / short a 125 put long a 135 call / short a 125 call

Bull and Bear Exercise (answers)

	<u>underlying price</u>	<u>directional outlook</u>	<u>implied volatility</u>	<u>spread choices</u>
d.)	248	bullish	unusually low	long a 250 call / short a 230 call <i>long a 250 call / short a 270 call</i> long a 250 put / short a 230 put long a 230 put / short a 250 put